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Computer Use for Neuromotor Paralysis Below the Neck Application

1. Introduction to our project and design

The goal of this lab final project was to design a human-computer interface to simulate cursor movement and clicks using an accelerometer and force sensor placed on the head. We needed to create a circuit and ensure clean, filtered, usable signals. We also needed to design and develop a LabVIEW program to ensure the implementation of the circuit into system movements. Finally our team ensured that the design was optimized and validated through physical use and completing a game level.

2. System Description and Overview

The system uses head tilt movements to move a computer cursor in the X and Y directions and a “bite force” gesture to execute mouse clicks. An accelerometer mounted on a small breadboard measures changes in orientation, which are translated into cursor displacement. A force sensor then detects force, which triggers a click event. Signals from both sensors pass through the myDAQ. LabVIEW processes the signals in real time and outputs cursor control commands to the computer from user32.dll.

3. Physical Circuit Design

As part of the team’s preliminary tasks and checks, the structure required to produce a successful signal through the utilization of the force sensor was conducted. Similar to the stretch sensor lab earlier in the semester, the team tested the components by designing a voltage divider circuit and observing the readout of a LabView code at 5V with three separate resistor values: 1k Ω , 10k Ω , 20k Ω . Using these three different resistor values in place of $R1$ of the voltage divider allowed the team to test the force sensor to find which resistor completed the circuit and provided the largest voltage difference between the sensor’s on and off states.

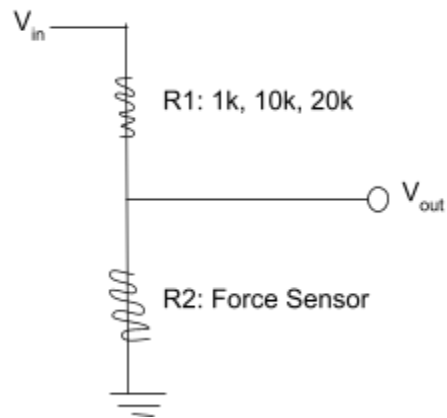


Figure 01: This figure is a circuit diagram representative of the voltage divider used to test the different resistor values with the force sensor, to determine which produces the most favorable voltage range. The model depicted here is the blueprint for the physical circuit build that was constructed at the time of testing in the lab. The values for $R1$ were tested individually in the order listed in the figure.

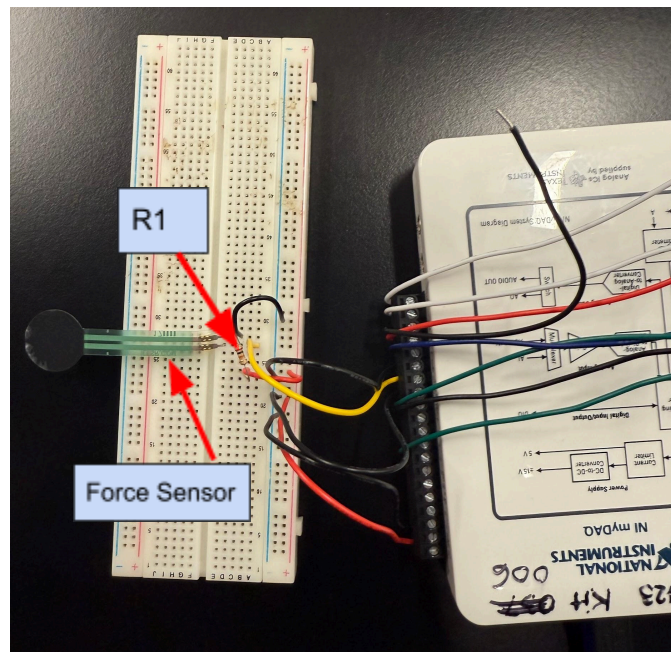


Figure 02: The figure depicts the physical test build that was constructed and used in lab. The force sensor and $R1$ (all three resistor values tested) can be seen in their respective locations to satisfy a tangible voltage divider build.

Resistor	Min V	Max V	Difference
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1K	3.4	5	1.6
10K	2	5	3
20K	1	5	4

Upon testing the three separate resistor values for $R1$, the one that holds a value of $20k\Omega$ produced the most favorable voltage range between the force sensors ‘off’ and ‘on’ states, sitting at around 4V. Therefore, the group chose the $20k\Omega$ resistor from the three, seeing that it displayed the largest difference.

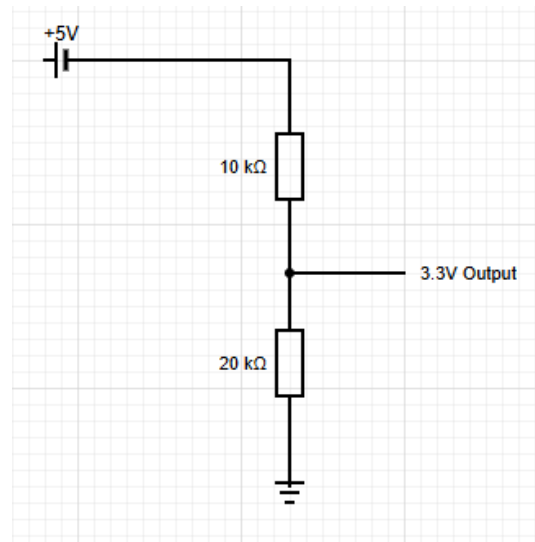


Figure 03: This figure is a circuit diagram representative of the voltage divider used to get the 5V signal from the myDAQ to the accelerometer at an input of 3.3V.

After choosing the correct component values, the motion sensor was connected with the MyDAQ, utilizing the same resistor values from the predetermined voltage divider. A LabView program was created connecting the motion sensor into the analog terminal of the MyDAQ. We then used a LPF in labview with a cutoff frequency of 5 Hz. This attenuated the high frequency and let our signal be cleaner and readable. Two waveform charts were connected to the filtered signal which displayed the x and y position of the accelerometer.

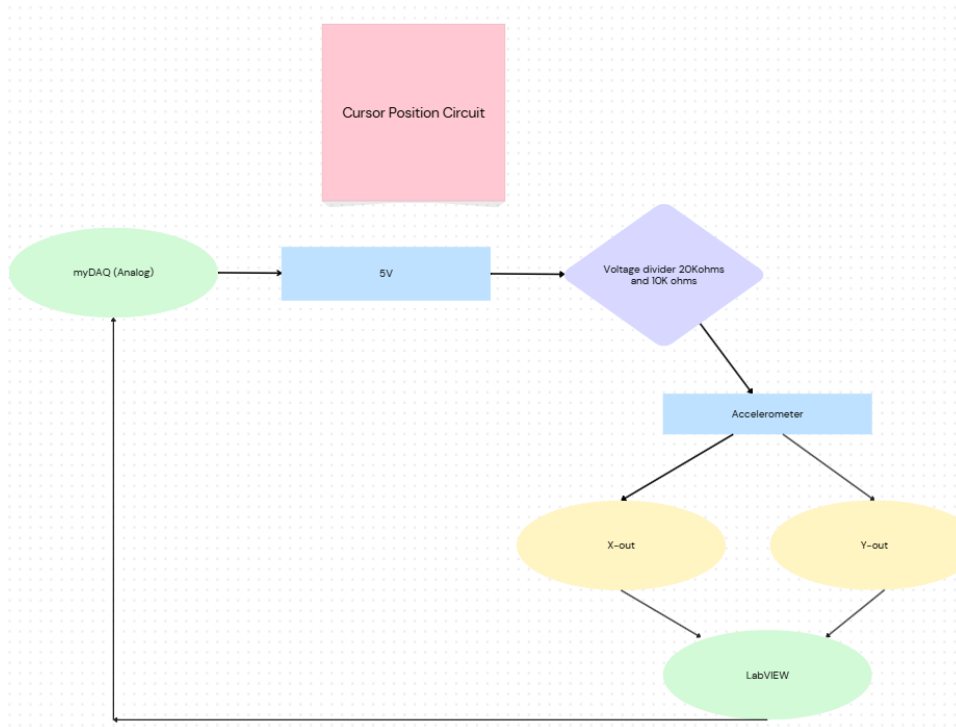


Figure 04: This figure shows the physical circuit process that we will use to connect our accelerometer to the myDAQ to further be used in adjusting cursor positions.

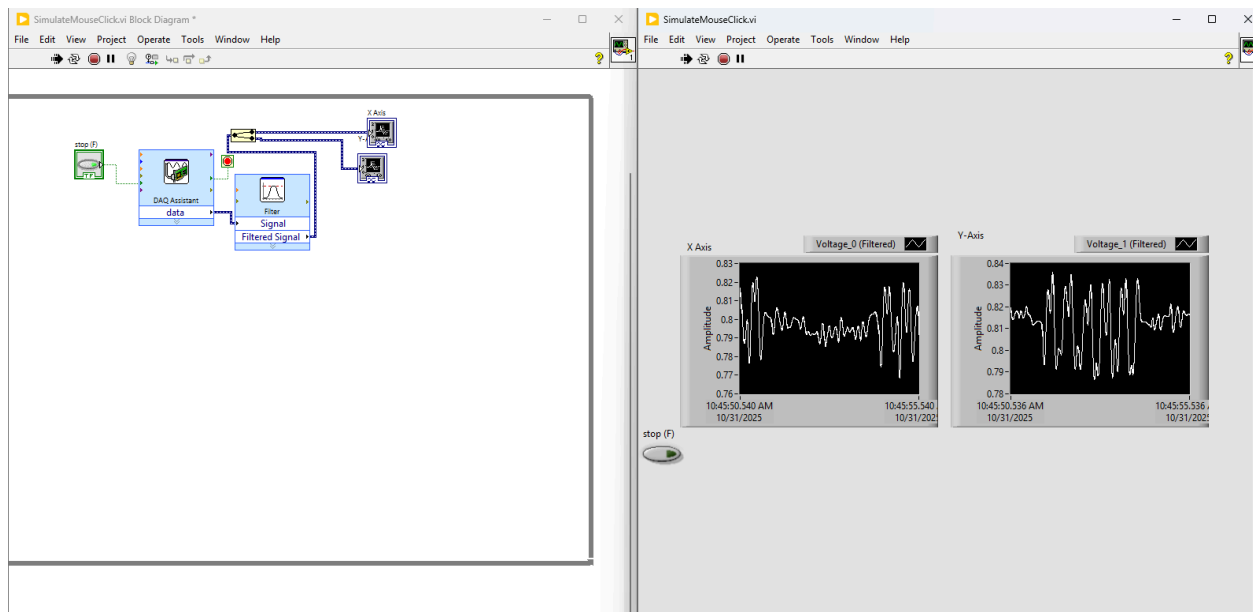


Figure 05: This figure shows the block diagram and front panel of LabVIEW. In an active while loop, the DAQ analog signal is taken from the accelerometer and passed into a low pass filter with a cutoff of 5 HZ. The filtered signal is then split into different waveform charts. Each chart shows the relative x and y positions of the accelerometer. We can see that the oscillations of the x axis and y axis have increased amplitudes at inverse times, this proves that our circuit setup was correct.

4. User Manual

4.1 User Setup

1. Mount accelerometer to headphones and ensure wire connection to myDAQ
2. Insert force sensor into mouth (Not for demonstration purposes)
3. Open LabVIEW VI
4. Run Block Diagram

4.2 Operating Instructions

1. Tilt head left/right -> Cursor X movement
2. Tilt head up/down -> Cursor Y movement
3. Bite gently -> Mouse click

4.3 Safety Notes

1. Do not bite excessively hard
2. Do not use jerky head movements
3. Secure wires before use
4. Take breaks to avoid neck and or jaw fatigue

4.4 Troubleshooting

1. Cursor jitter -> verify filter values and front panel waveform charts
2. No click detected -> decrease force threshold.
3. No cursor movement -> check accelerometer and circuit wiring.

5. Cursor Control

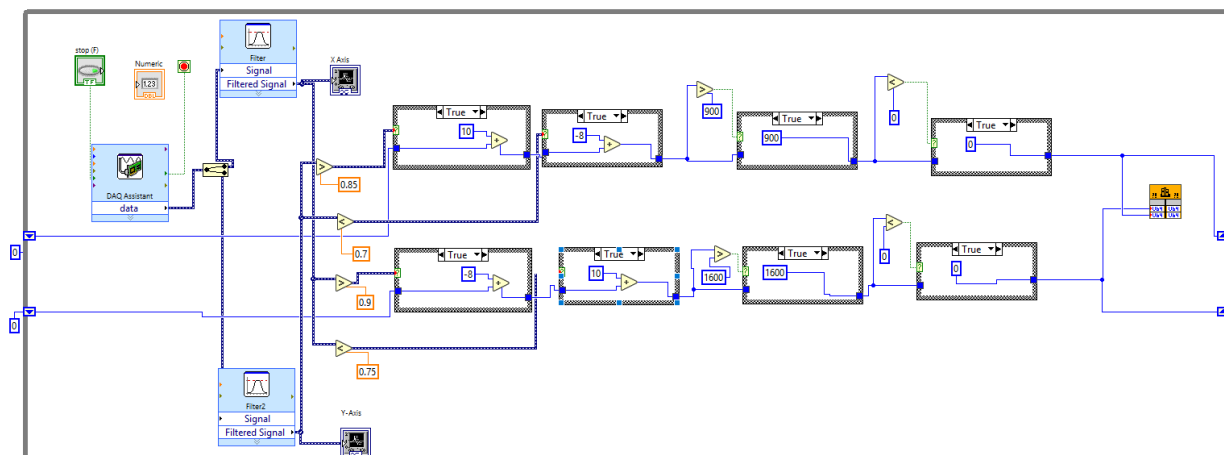


Figure 06: The cursor control diagram is utilizing the analog DAQ assistant that is filtered and split into an x and y axis as described previously, encompassed in a while loop. For the x axis, there are 4 for loops, where the cases are as follows: the signal is measured and if the amplitude is greater than 0.85, the mouse cursor moves 10 points to the right, if it is less than 0.7 it moves 8 points to the left. These are the first two for loops. After that, the next two for loops determine if the mouse cursor has moved to ± 900 on the x axis, if so the cursor stays and does not continue to move to the right or left, this is to avoid the cursor moving off of the monitor which has a resolution of 16:9. The same loop is repeated for the y axis, but with slightly adjusted frequency values, these are all input into the call

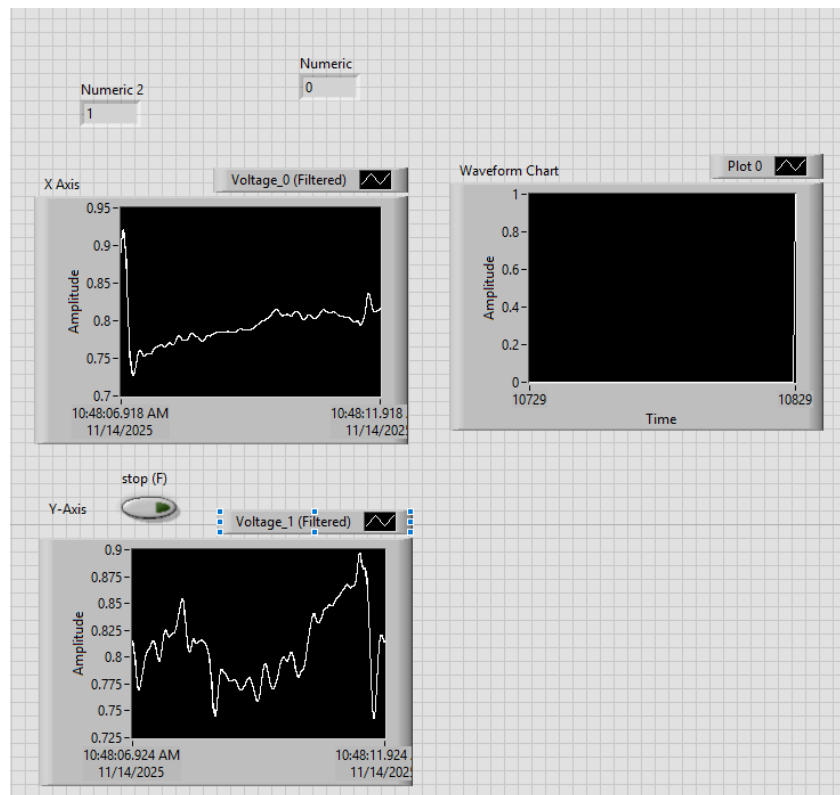
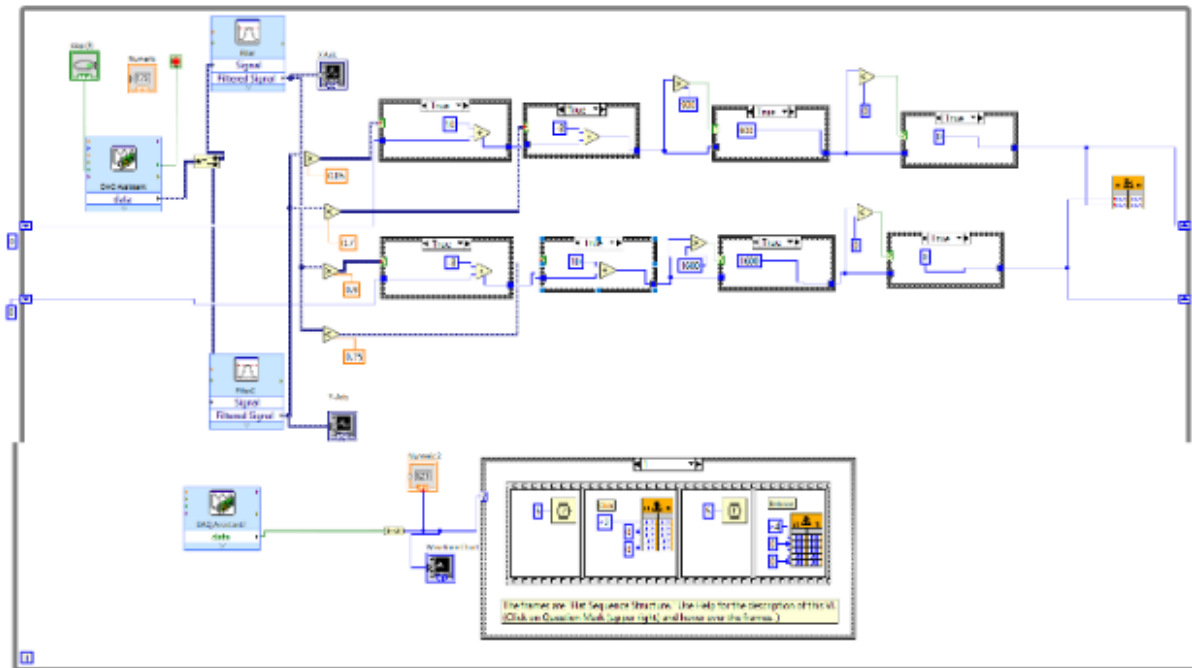


Figure 9: This is the completed LabVIEW block diagram and front panel from the mouth controlled computer mouse. The x and y axis waveform charts show the recent mouse movements in each direction. The right side waveform chart displays the force sensor and whether a click has been registered. We can see that at the most recent moment, the waveform chart did indeed display a click.

8. Experiment Design and Optimization

For experimental evaluation, we mounted the device onto a pair of headphones worn by the lovely test user, Tyler. After launching the LabVIEW interface, Tyler used head movements to control the cursor and successfully navigated to a game of tic tac toe on the Google browser. Using only the accelerometer for cursor control and the force sensor for clicking, he proceeded to absolutely wipe the floor with the bot and catch the win.

Following the free win, Tyler navigated back to the LabVIEW application and turned off the system, once again using only the head mounted accelerometer device. This demonstration confirmed complete operational functionality of both the cursor movement and clicks.

To further optimize performance, we adjusted the system's cursor movement speed by increasing the value in the for loops of each axis. Tyler reported that the initial configuration felt sluggish and required excessive head movement and patience. Increasing sensitivity improved responsiveness and reduced the effort required for navigation. After tuning the sensitivity, overall operation felt significantly smoother and more natural. Based on this optimization and demonstration, Amulya approved the system as meeting requirements.

9. Final Performance and System Analysis

9.1 Cursor Movement

Smoothness: Excellent
Response Time: Excellent
Full range of motion: Excellent

9.2 Click Performance

Reliability: Excellent
Response Time: Excellent

9.3 User Feedback

The design was intuitive and relatively comfortable. I wish the wires were longer and soldered to the board.

10. Conclusion

The lab was successful and a functional assistive mouse using head movements was created. Real-time signal processing, circuit design, and design optimization were used. Further

improvements could include soldered connections, improved comfort and possible right-click capabilities.